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Stay Focused! Exploring the Compulsive Nature of Alcohol-Related Attentional Bias in Severe Alcohol Use Disorder

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Objective: Prominent models postulate that alcohol-related attentional bias (AB), emerging from the overactivation of the reward system, plays a key role in severe alcohol use disorder (sAUD) and is independent from voluntary control. We determined whether AB is indeed compulsive or can be modulated by the control/inhibition system. ***Method:*** Patients with sAUD (17 women, 13 men, mean age of 47, White) and matched healthy controls (16 women, 14 men, mean age of 44, White) performed a visual probe task with behavioral (reaction time) and eye-tracking (first fixation location and duration, second fixation location, dwell time) measures. They also performed an avoidance task, requiring to focus on a target by voluntarily inhibiting eye movements toward alcohol/nonalcohol/nonappetitive distractors and measuring overt (break frequency) and covert (fixational eye movements) attentional processes. ***Results:*** Patients with sAUD exhibited an avoidance AB indexed by (a) reduced attentional resources dedicated to alcohol-related stimuli, namely, reduced dwell time ($p = .040$) and second fixation ($p = .001$) toward these stimuli; (b) increased inhibitory processes, namely, easier inhibition of saccades toward alcohol measured by lower break frequency ($p < .001$); and (c) covert eye movements posited further away from alcohol. ***Conclusions:*** In contradiction with theoretical models, our two tasks did not show any AB toward alcohol in sAUD. Instead, patients exhibited an avoidance AB indexed by increased inhibitory processes as well as reduced overt and covert attentional resources dedicated to alcohol-related stimuli. These results question the theoretical and clinical role of AB, as measured through reliable eye-tracking tasks, in sAUD.

Public Health Significance Statement

Detoxified patients with severe alcohol use disorder are not characterized by a compulsive and irrepressible hijacking of attentional resources toward alcohol-related stimuli, as stated by dominant models, but rather avoid processing them, as shown here in two tasks measuring attentional bias toward alcohol.

Keywords: attentional bias, alcohol use disorder, eye tracking, inhibitory control, compulsivity

Severe alcohol use disorder (sAUD) is a key public health concern (Rehm & Shield, 2019). Numerous studies have underlined its damaging consequences on cognitive (Stavro et al., 2013) and cerebral (Bühler & Mann, 2011) functioning, and it is therefore crucial to better understand the key mechanisms involved in its emergence and persistence. Alcohol-related attentional bias (AB), reflecting the tendency to orient attentional resources toward alcohol-related stimuli, may constitute one such mechanism. Indeed, most theoretical frameworks postulate that AB is a central feature of sAUD. For example, dual-process models (Bechara, 2005; Wiers et al., 2007) suggest that sAUD would rely on altered interactions between two brain systems: The reflective/control system, responsible for high-level functioning and deliberative behavior, becomes underactivated (through the neurotoxic effects of excessive alcohol consumption, notably on frontal structures), while the reflexive/reward system, involved in the appetitive evaluation of stimuli, becomes overactivated (by the repeated reward emerging from the exposure to alcohol-related cues). Such dysregulation

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results in reduced executive abilities among patients with sAUD (Stavro et al., 2013) and increased attention toward alcohol-related cues, namely, AB (considered as indexing the overactivation of the reflexive/reward system).

However, the key role of AB in sAUD has been questioned: Most experimental studies did not document AB in recently detoxified patients with sAUD (Bollen et al., 2022 for a review), some even reporting an alcohol avoidance AB (Fridrici et al., 2013; Townshend & Duka, 2007). An eye-tracking study notably showed lower dwell time and less second fixations (indexing lower attentional reengagement) on alcohol-related stimuli in a visual probe task (VPT) in sAUD (Bollen et al., 2021). These findings question the proposal that AB is a stable component of sAUD and rather point to the influence of current drinking or clinical context on AB magnitude. Indeed, there is a massive gap between the sAUD population described by traditional models (i.e., individuals with current alcohol consumption) and the sAUD population included in most experimental studies (i.e., recently detoxified patients under treatment), which might explain the discrepancy between the pattern of AB expected by models and the one experimentally reported. Some studies showed that AB in sAUD would be highly affected by current craving or drinking status (Bollen et al., 2024; Field et al., 2013; Sinclair et al., 2016). This supports theoretical and experimental proposals that AB is driven by temporary changes in motivation toward alcohol, hence reflecting the momentary subjective evaluation of alcohol-related stimuli (Field et al., 2016; Noël et al., 2006). Most patients tested in previous studies were abstinent and undergoing detoxification treatment. Such states are frequently related to aversive or ambivalent alcohol evaluations, which could explain the inconsistencies across previous work exploring AB without measuring the motivational state toward alcohol at testing time.

A remaining question is whether this modulation of AB by motivational states is compulsive and driven automatically or whether it requires the activity of the reflective/control system (e.g., inhibitory control) to direct attention toward stimuli congruent with these states. Traditional AB paradigms do not disentangle the involvement of automatic versus controlled processes. For example, the VPT consists in free visual exploration without instructions. In the same vein, while mobilizing inhibition abilities, the addiction Stroop task does not allow to determine AB direction. Indeed, higher Stroop interferences for alcohol-related words could result from either the automatic allocation of attention to the semantic processing of alcohol-related words, or conversely from an attempt to avoid processing these words (Klein, 2007). Moreover, both paradigms rely on reaction times to assess AB, such measures showing very poor reliability (Ataya et al., 2012).

An eye-tracking avoidance task with a gaze contingency procedure was recently developed to explore the direct control of reflective abilities on early saccadic movements toward alcohol-related cues, by measuring the ability to inhibit the orientation of attention toward alcohol-related distractors when focusing on a specified target (Qureshi et al., 2019; Wilcockson & Pothos, 2015). This task requires the deliberate inhibition of AB (i.e., the production of early saccadic movements toward alcohol-related stimuli), thus testing the ability of reflective processes to modulate alcohol-related AB and hence clarifying the automatic/controlled nature of AB. This paradigm showed that heavy/problem drinkers were more likely to break their central focus and produce saccadic eye movements toward alcohol-related distractors compared to neutral nonappetitive distractors (the frequency with which gaze fixations were made on distractors being termed "break frequency"; Qureshi et al., 2019; Wilcockson & Pothos, 2015). Qureshi et al. (2019) further demonstrated that these findings may be partly attributable to the appetitive content of the stimuli, as higher break frequency generalized toward other appetitive stimuli (i.e., nonalcohol beverages). They also showed that higher break frequency for appetitive stimuli was only observed when stimuli appeared in peripheral vision (as opposed to stimuli adjacent to the target), which they explained by the impossibility to recruit covert attention to process remote stimuli. However, one may argue that participants would exhibit more random than content-specific AB when they are unable to attentionally process distractors.

While promising, this paradigm still needs to be validated in a clinical population of patients with sAUD, notably by comparing the "attentional break" frequency measures with reliable and classical AB measures as the VPT. While both tasks assess alcohol-related AB, they explore it under different conditions (free exploration vs. forced avoidance) and might thus tap into distinct attentional processes (attentional monitoring vs. inhibitory processes). The recruitment of covert attention to process nearby versus remote distractors could also be investigated through innovative eye-tracking measures. Indeed, eyes are in constant motion even when the gaze is maintained on a fixated point, resulting in micro eye movements like tremors, drifts, and microsaccades (Engbert, 2006; Hafed et al., 2015). Among those fixational eye movements, microsaccades direction has been widely claimed to reflect shifts in covert attention (Laubrock et al., 2007; Lv et al., 2022; Yuval-Greenberg et al., 2014).

We therefore explored the inhibitory and controlled processes of AB in sAUD by applying the paradigm developed by Qureshi et al. (2019) in a population of recently detoxified patients with sAUD. Moreover, we explored the convergent validity of this novel paradigm with the eye-tracking VPT. We also determined the impact of current motivational state (i.e., subjective craving at testing time) on AB measures and whether break frequency could be affected by the predictability of the type of stimuli presented. To do so, we also used a predictable version of this avoidance task in which we separated the trials in three blocks according to the type of distractors (alcohol-related, nonalcohol, nonappetitive) and informed participants about the type of distractors that will appear. Finally, we conducted exploratory analyses to track putative covert AB that we measured by tracking gaze position of fixational eye movements (i.e., when participants maintained their gaze on the target).

Method

Participants

We recruited 30 inpatients (17 women, 13 men, White), fulfilling the *Diagnostic and Statistical Manual of Mental Disorders, fifth edition*, criteria for sAUD, during their detoxification treatment in Belgian hospitals. They had all abstained from alcohol for at least 7 days, were free of psychiatric comorbidities (except tobacco use disorder), and were not involved in any attentional training program. We matched patients for age and gender with 30 control participants (CTL; 16 women, 14 men, White). CTL were free of any past or present psychiatric disorder as well as any personal or parental history of sAUD. They consumed less than 10 alcohol units per week and never exceeded three units per day. They scored below the cutoff score of 8 on the Alcohol Use Disorders Identification Test (Babor & Robaina, 2016). Exclusion criteria for both groups included polysubstance use disorder and major past or present neurological disorder and/or trauma. All participants had normal or corrected vision and were fluent French speakers. We determined our sample size based on previous eye-tracking studies that detected medium-to-large effect sizes for interactions indexing the presence of alcohol-related AB (Bollen et al., 2021, 2024; Qureshi et al., 2019).

Procedure

Participants provided written informed consent, were not aware of the hypotheses, and were tested individually in a quiet room. They first completed questionnaires assessing state anxiety (State Trait Anxiety Inventory–A; Bruchon-Schweitzer & Paulhan, 1993) and current alcohol craving (Alcohol Craving Questionnaire–Short Form–Revised [ACQ-SF-R] and Visual Analogue Scale [VAS]: "Indicate how much you want to drink alcohol right now [from 0 = *not at all*, to 100 = *strongest desire*]") before performing the experimental tasks. Participants performed the tasks in the following order: VPT, mixed version of the avoidance task (replicated from Qureshi et al., 2019) and predictable version of the avoidance task. Each task lasted about 15 min. Between tasks, participants completed questionnaires assessing depressive symptoms (Beck Depression Inventory–II; Beck et al., 1996), trait anxiety (State Trait Anxiety Inventory–B; Bruchon-Schweitzer & Paulhan, 1993), and impulsivity (UPPS-P Impulsive Behavior Scale; Billieux et al., 2012). The study was performed in accordance with the ethical standards of the Declaration of Helsinki and was approved by the Ethics Committee of the Saint-Luc-UCLouvain Clinics (reference number: 2019/26MAR/141). At the end of the experiment, participants were debriefed, and CTL received financial compensation.

Apparatus

Participants seated 60 cm away from a laptop, facing an eye-tracker camera and an Asus Display Laptop PC with a 17.3-inches screen (resolution 1080 × 1920 pixels; refresh rate 120 Hz). We controlled the presentation of the experimental tasks and their synchronization with OpenSesame (Mathôt et al., 2012) and Experiment Builder (SR Research Ltd.). We recorded eye movements using an EyeLink Portable Duo remote mode eye tracker (SR Research, Canada; sampling rate of 1000 Hz; average accuracy range 0.25°–0.5°, gaze tracking range of 32° horizontally and 25° vertically). We performed a 9-point calibration of participant's eye position at the beginning of each block of the tasks.

VPT

For each trial, a central fixation dot first appeared on a black background, and participants had to fixate their gaze on it. Once the eyes were detected at the center of the screen by the eye-tracking device, the fixation dot was replaced by the onset of two pictures (i.e., alcohol beverage picture and soft drink beverage picture). They were displayed in a counterbalanced order on the left/right side of the computer screen for 2,000 ms and then replaced by a white arrow probe appearing at the location previously occupied by one of the pictures. Participants had to report the orientation of the probe by pressing the "up" or "down" key on the keyboard, as quickly and correctly as possible. Visual probes replaced the two types of pictures with equal frequency. Each trial was separated by an intertrial interval of random duration (500–1,500 ms). The task contained 84 trials, including four practice trials excluded from the analyses.

Twenty pairs of alcohol beverage pictures (e.g., bottle of vodka, can of beer) and matched nonalcohol beverage pictures (e.g., bottle of water, soft drink can) without context were extracted from the Alcohol Beverage Picture Set battery (Pronk et al., 2015). The visible brands and writings of the beverage were blurred to avoid reading. Each picture pair was matched on the following perceptual features: color, size (444 × 444 pixels or 10.7° × 10.7° angle), brightness, and salience using the SHINE toolbox (Willenbockel et al., 2010). We presented each pair of stimuli 4 times (2 Stimuli Position × 2 Probe Position).

We assessed performance through behavioral and eye-tracking measures. The *reaction time* for probes replacing the stimulus (alcohol or nonalcohol) is the traditional behavioral AB index (in ms). The *first fixation location* indicates whether a stimulus was first fixated (i.e., initial attentional capture; dummy coded as 1 if fixated first, 0 if not fixated first). The *first fixation duration* indicates the duration of this first fixation (i.e., persistence of attentional focus; in ms). The *second fixation location* indicates whether the the stimulus not fixated first was fixated second (i.e., attentional switch; dummy coded as 1 if fixated second, 0 if not). The *dwell time* is the sum of fixation times on each stimulus during the whole trial (i.e., attentional maintenance; in ms). We qualified gaze samples as fixations or saccades according to the standard EyeLink algorithms.

Avoidance Task

We replicated the gaze contingency paradigm from Qureshi et al. (2019), using three categories of alcohol appetitive, nonalcohol appetitive, and nonalcohol nonappetitive visual cues. Participants had to maintain their gaze on a fixation target and ignore the distractor. Target and distractor were randomly presented within one of the nine regions on the screen. Half of the trials presented a nearby distractor (distance between stimuli centers: 209–385 pixels) and the other half a remote distractor (distance: 514–784 pixels). For each trial, the distractor appeared on the screen once participants gazed at the target for 1,000 ms. If participants performed a saccade away from the target, the distractor disappeared and only reappeared when they returned their gaze to the fixation target for 10 ms. For each trial, the fixation target was displayed for 5,000 ms, the maximum onscreen time for the distractor being 4,000 ms.

We used 90 images as distractor stimuli, with 30 alcohol appetitive stimuli, 30 nonalcohol appetitive stimuli, and 30 nonappetitive stimuli. All distractor stimuli had the same size (225 × 225 pixels), and the resolution of the computer screen was set to 1280 × 1024 to use the same visual angle than Qureshi et al. (2019). They were matched on valence, arousal, angles of objects, luminance, and color. Pictures of alcohol and nonalcohol beverages were extracted from the Alcohol Beverage Picture Set (Pronk et al., 2015) and were matched with nonappetitive products (e.g., fabric softener, cleaning items) from Qureshi et al. (2019).

We presented 91 trials (30 per category of distractors and one initial blank distractor) in each version of the task. In the mixed version, we randomized the distractor stimuli to provide a strict replication of Qureshi et al. (2019). In the predictable version, we dissociated trials based on the distractor type and presented them sequentially (the order of the blocks and of the 30 trials within each block being randomized). We informed participants about the distractor type before starting each block.

We assessed performance through a single eye-tracking index: the *break frequency*. The break indicates whether the participant performed, during the trial, any saccade outside the target area and directed toward the distractor (i.e., inhibitory control failure; dummy coded as 1 if yes, 0 if no).

Data Reduction and Statistical Analyses

We performed a data reduction procedure on reaction times for the VPT. Before the analysis, we removed trials with incorrect responses (0.039% of trials) and with reaction times higher than 2,000 ms (0.003% of trials). No trial with reaction times lower than 200 ms was recorded. All participants performed the VPT and the mixed avoidance task (N = 60), but we missed data in some questionnaires (7 sAUD, 1 CTL) and in some blocks of the predictable avoidance task (3 sAUD, 2 CTL) due to participants dropout, data loss, or poor eye-tracking calibration.

To measure task reliability, we computed AB scores for *reaction time* (reaction time for probes following nonalcohol minus probes following alcohol), *first fixation location* (proportion of first fixation on alcohol compared to nonalcohol), *first fixation duration* (duration of first fixations on alcohol minus nonalcohol), and *dwell time* (dwell time on alcohol minus nonalcohol). A positive/negative AB score for reaction time, first fixation duration, and dwell time and a score over/under 50% for first fixation location indicated AB toward/away from alcohol. We also measured the reliability of the *second fixation location* (i.e., proportion of second fixation on alcohol/nonalcohol compared to no fixation when the first fixation was on nonalcohol/alcohol in the VPT) and the *break frequency* (i.e., proportion of trials in which gaze fixations were made toward the alcohol/nonalcohol/nonappetitive distractor in the avoidance task) for each type of stimuli, as these proportions were based on separate trials and did not inform about AB for the other type(s) of stimuli. To perform correlational analyses on AB measurements in the avoidance task, we computed AB scores for break frequency comparing alcohol with nonalcohol (break frequency on alcohol minus nonalcohol) or nonappetitive stimuli (break frequency on alcohol minus nonappetitive). We performed all statistical analyses using the SPSS 27.0 software.

We performed between-group comparisons (i.e., independent t tests) on demographic, alcohol-related, and psychopathological variables. For the VPT, we indexed the internal consistency of the task by computing Cronbach's α on all AB scores for the 20 pairs of pictures for each participant. We then performed repeated-measures analyses of variance on behavioral (reaction time) and eye-tracking (first fixation location, first fixation duration, second fixation location, dwell time) measurements with GROUP (sAUD, CTL) as between-subjects factor and TYPE (alcohol, nonalcohol) as within-subjects factor. For the avoidance task, we computed Cronbach's α for the 30 trials of each version of the task and for each participant. We performed analyses of variance for each version of the task on break frequency with GROUP (sAUD, CTL) as between-subjects factor and TYPE (alcohol, nonalcohol, nonappetitive) and DISTANCE (nearby, remote) as within-subject factors. We conducted post hoc tests (independent samples t tests) with a Bonferroni correction of α/2 = .025. We performed Pearson correlations to explore the convergent validity of the two AB tasks and nonparametric Spearman correlations to explore their association with craving (to account for the large number of participants reporting a craving score of 0).

For our exploratory analyses on fixational eye movements, we split the total presentation time of distractors (5,000 ms) into 50 ms bins for each trial and then calculated the distance (in degrees) between gaze position and target center in each bin (positive/negative values indexed fixational eye movements directed toward/away from distractor). We excluded trials in which gaze position was located outside the target (i.e., attentional break trials) by removing distances higher than target radius. For each remaining trial, we centered our measures on the distance between eye position and target center 100 ms before distractor onset. We performed independent and paired samples t tests on each bin to explore the effects of DISTANCE (nearby, remote), TYPE (alcohol, nonalcohol, nonappetitive) and GROUP (sAUD, CTL). In these exploratory analyses, we used a more restrictive p-value threshold (.010) and reported significant differences only when they constituted a coherent pattern occurring in minimum five consecutive bins.

Transparency and Openness

We report how we determined all data exclusions, all manipulations, and all measures in the study. The data set and analysis code associated with this study are openly available on the Open Science Framework at https://osf.io/z8xqa. This study design and analyses were not preregistered.

Results

Demographic, Psychological, and Alcohol-Related Measures (Table 1)

Patients with sAUD reported higher scores than CTL on Alcohol Use Disorders Identification Test, depression, trait anxiety, and impulsivity measures. Groups did not differ regarding gender, age, state anxiety, and craving (assessed through ACQ or VAS).

VPT

Reliability Estimates

Internal consistency was under the 0.70 cutoff score of acceptable internal reliability (Kline, 2000) for reaction time (α = .517), first fixation location (α = .221), and duration (α = .507) measures. Conversely, it was excellent for second fixation location measures (alcohol: α = .967; nonalcohol: α = .906) and dwell time (α = .981).

Reaction Times

We found a GROUP main effect, $F(1, 58) = 12.357$, $p < .001$, $\eta_p^2 = .176$, showing longer reaction times for patients with sAUD (alcohol: 736 ± 194 ms; nonalcohol: 751 ± 209 ms) than CTL (alcohol: 589 ± 133 ms; nonalcohol: 591 ± 129 ms). We also showed a TYPE main effect, $F(1, 58) = 4.234$, $p = .044$, $\eta_p^2 = .068$,

Table 1

Demographic, Psychological, and Alcohol-Related Measures (Means ± Standard Deviations) of Patients With sAUD and CTL

Measure	sAUD (N = 30)	CTL (N = 30)	t or χ^2	p	d or ϕ
Demographic measures					
Gender ratio (women/men)	17/13	16/14	0.067	.795	0.033
Age	47.23 ± 9.63	44.07 ± 11.71	1.144	.257	0.295
Psychopathological measures					
Beck Depression Inventory	9.54 ± 4.86[7]	2.47 ± 3.05	**6.120**	<.001	1.799
State Anxiety Inventory	34.38 ± 10.36	30.83 ± 10.36	1.327	.190	0.343
Trait Anxiety Inventory	51.30 ± 10.18[6]	35.23 ± 9.74	**5.904**	<.001	1.617
Impulsivity	45.30 ± 9.19[7]	38.72 ± 6.26[1]	**2.934**	.006	0.855
Alcohol consumption measures					
AUDIT	28.70 ± 7.07[7]	3.33 ± 1.84	**16.779**	<.001	5.234
Craving (VAS)	7.83 ± 21.92	4.50 ± 11.68	0.735	.465	0.190
Craving (ACQ)	21.75 ± 9.53	20.63 ± 8.79	0.472	.639	0.122
Number of *DSM-5* criteria	7.93 ± 1.93	N/A			
sAUD duration (in years)	9.88 ± 9.67[1]	N/A			
Previous detoxification stays	1.79 ± 2.27[1]	N/A			

Note. The superordinate numbers indicate the number of missing observations. Values in bold indicate significant results. sAUD = severe alcohol use disorder; CTL = control participants; AUDIT = Alcohol Use Disorders Identification Test; VAS = Visual Analogue Scale; ACQ = Alcohol Craving Questionnaire; *DSM-5* = *Diagnostic and Statistical Manual of Mental Disorders, fifth edition*; N/A = not applicable to this group.

showing faster reaction times for alcohol compared to nonalcohol stimuli, but no GROUP × TYPE interaction, $F(1, 58) = 2.315$, $p = .134$, $\eta_p^2 = .038$.

Early Processing Stages

First Fixation Location. We found no significant GROUP main effect, $F(1, 58) = 0.305$, $p = .583$, $\eta_p^2 = .005$, TYPE main effect, $F(1, 58) = 0.564$, $p = .456$, $\eta_p^2 = .010$, nor GROUP × TYPE interaction, $F(1, 58) = 0.767$, $p = .385$, $\eta_p^2 = .013$ (sAUD: alcohol: 48.25% ± 7.00%; nonalcohol: 46.19% ± 8.02%; CTL: alcohol: 47.82% ± 5.65%; nonalcohol: 47.98% ± 6.38%).

First Fixation Duration. We found no significant GROUP main effect, $F(1, 58) = 3.032$, $p = .087$, $\eta_p^2 = .050$, TYPE main effect, $F(1, 58) = 3.334$, $p = .073$, $\eta_p^2 = .054$, nor GROUP × TYPE interaction, $F(1, 58) = 0.312$, $p = .578$, $\eta_p^2 = .005$ (sAUD: alcohol: 261 ± 73 ms; nonalcohol: 280 ± 81 ms; CTL: alcohol: 240 ± 48 ms; nonalcohol: 250 ± 55 ms).

Late Processing Stages

Second Fixation Location. We found a GROUP main effect, $F(1, 58) = 16.274$, $p < .001$, $\eta_p^2 = .219$, showing a reduced proportion of second fixation (vs. no fixation) in sAUD (alcohol: 54.62% ± 33.23%; nonalcohol: 76.69% ± 29.37%) compared to CTL (alcohol: 87.54% ± 13.73%; nonalcohol: 87.54% ± 15.61%), and a TYPE main effect, $F(1, 58) = 11.534$, $p = .001$, $\eta_p^2 = .166$, showing a reduced proportion of second fixation toward alcohol compared to nonalcohol stimuli. However, the most important result was the GROUP × TYPE interaction, $F(1, 58) = 11.530$, $p = .001$, $\eta_p^2 = .166$, qualifying these main effects: After having first fixated a nonalcohol stimulus, patients with sAUD were less likely than CTL to perform a second fixation (vs. no fixation) toward alcohol, $t(38.629) = 5.015$, $p < .001$, Cohen's $d = 1.295$, while groups did not differ regarding second fixation toward nonalcohol stimuli, $t(44.173) = 1.786$, $p = .081$, Cohen's $d = .461$. Moreover, patients with sAUD showed a lower proportion of second fixation for alcohol compared to nonalcohol, $t(29) = 3.601$, $p = .001$, $d = .657$, while this difference was not found in CTL, $t(29) = 0.001$, $p = .999$, $d = .000$. Figure 1A offers a graphical illustration of these results and presents individual data.

Dwell Time. We found no GROUP main effect, $F(1, 58) = 0.152$, $p = .698$, $\eta_p^2 = .003$, but a TYPE main effect, $F(1, 58) = 14.133$, $p < .001$, $\eta_p^2 = .196$, dwell time being longer for nonalcohol (sAUD: 700 ± 409 ms; CTL: 611 ± 215 ms) than for alcohol stimuli (sAUD: 398 ± 238 ms; CTL: 525 ± 175 ms). However, the most important result was the GROUP × TYPE interaction, $F(1, 58) = 4.430$, $p = .040$, $\eta_p^2 = .071$, showing that patients with sAUD presented shorter dwell times on alcohol than CTL, $t(58) = 2.362$, $p = .022$, $d = .610$, while groups did not differ regarding dwell times on nonalcohol, $t(43.899) = 1.063$, $p = .294$, $d = .274$. Moreover, patients with sAUD showed shorter dwell times on alcohol compared to nonalcohol, $t(29) = 3.204$, $p = .003$, $d = .585$, while this difference was not found in CTL, $t(29) = 2.053$, $p = .049$, $d = .375$. Figure 1B offers a graphical illustration of these results and presents individual data.

Avoidance Task

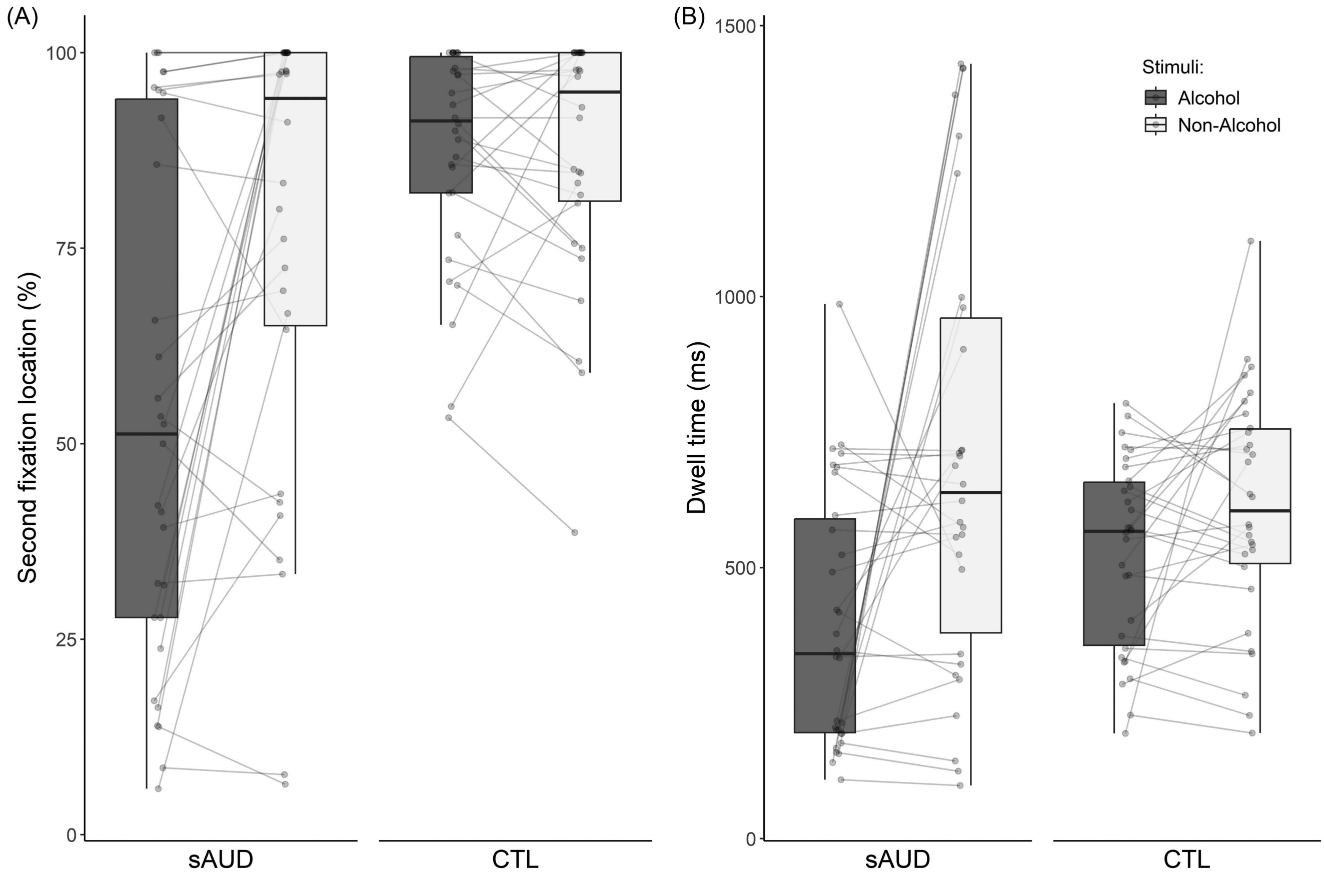
Reliability Estimates

Internal consistency was high for break frequency toward alcohol (mixed task: $\alpha = .895$; predictable task: $\alpha = .853$), nonalcohol (mixed task: $\alpha = .891$; predictable task: $\alpha = .878$), and nonappetitive (mixed task: $\alpha = .909$; predictable task: $\alpha = .905$) distractors.

Overt Attention–Break Frequency

Mixed Avoidance Task. We found a main effect of DISTANCE, $F(1, 58) = 66.650$, $p < .001$, $\eta_p^2 = .535$, showing higher break frequency in trials presenting nearby distractors (sAUD: alcohol: 0.36 ± 0.27; nonalcohol: 0.37 ± 0.27; nonappetitive: 0.37 ± 0.27/ CTL: alcohol: 0.24 ± 0.24; nonalcohol: 0.26 ± 0.23; nonappetitive: 0.28 ± 0.25) compared to remote ones (sAUD: alcohol: 0.21 ± 0.20;

Figure 1
Proportion of Second Fixations and Dwell Times in the Visual Probe Task



Note. The boxplots compare (a) the proportion of second fixations and (b) dwell times in detoxified patients with sAUD and CTL for alcohol and nonalcohol stimuli in the visual probe task. Dots indicate individual data, connected through gray lines. sAUD = severe alcohol use disorder; CTL = control participants.

nonalcohol: 0.19 ± 0.18; nonappetitive: 0.22 ± 0.23/CTL: alcohol: 0.13 ± 0.14; nonalcohol: 0.10 ± 0.16; nonappetitive: 0.14 ± 0.19). Main effects of TYPE, $F(2, 116) = 1.768$, $p = .175$, $\eta_p^2 = .030$, GROUP $F(1, 58) = 3.687$, $p = .060$, $\eta_p^2 = .060$, and interactions $F \leq .965$, $p \geq .384$, $\eta_p^2 \leq .016$ were not significant. Figure 2A offers a graphical illustration of these results and presents individual data.

Predictable Avoidance Task. We found a main effect of DISTANCE, $F(1, 53) = 50.499$, $p < .001$, $\eta_p^2 = .488$, showing higher break frequency in trials with nearby distractors (sAUD: alcohol: 0.26 ± 0.22; nonalcohol: 0.34 ± 0.25; nonappetitive: 0.37 ± 0.25/CTL: alcohol: 0.23 ± 0.22; nonalcohol: 0.23 ± 0.21; nonappetitive: 0.28 ± 0.24) compared to remote ones (sAUD: alcohol: 0.16 ± 0.16; nonalcohol: 0.19 ± 0.17; nonappetitive: 0.24 ± 0.21/CTL: alcohol: 0.09 ± 0.13; nonalcohol: 0.12 ± 0.17; nonappetitive: 0.14 ± 0.18). We also found a main effect of TYPE, $F(2, 106) = 9.065$, $p < .001$, $\eta_p^2 = .146$, showing lower break frequency for alcohol stimuli compared to nonalcohol and nonappetitive stimuli. Main effect of GROUP $F(1, 53) = 2.980$, $p = .090$, $\eta_p^2 = .053$ and interactions $F \leq 1.092$, $p \geq .339$, $\eta_p^2 \leq .020$ were not significant. Figure 2B offers a graphical illustration of these results and presents individual data.

Covert Attention–Fixational Eye Movements

Mixed Avoidance Task (Figure 3). We found a DISTANCE effect, showing that fixational eye movements were positioned closer to nearby than remote distractors, from 200 to 500 ms after distractor onset (6 consecutive 50 ms bins with $p < .01$).

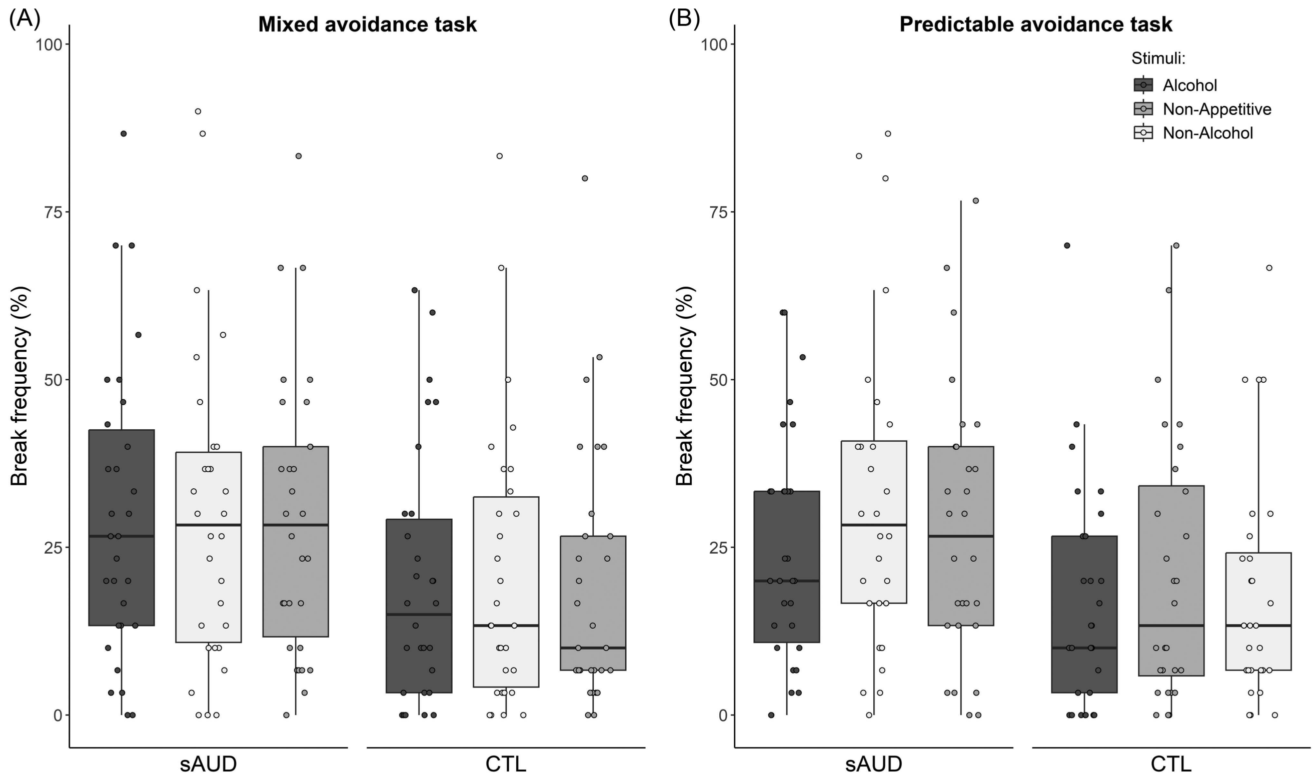
Predictable Avoidance Task (Figure 4). We found a DISTANCE effect, showing that fixational eye movements were positioned closer to nearby than remote distractors, from 250 to 600 ms after distractor onset (7 bins with $p < .01$). When focusing on nearby distractors, we showed that fixational eye movements of patients with sAUD were positioned farther away from alcohol distractors compared to nonappetitive distractors, from 2,650 to 2,900 ms (5 bins) and from 2,950 to 3,200 ms (5 bins) and compared to nonalcohol distractors, from 3,600 to 3,850 ms (5 bins).

Correlational Analyses

Convergent Validity

In the whole sample, dwell time AB score in the VPT did not correlate with break frequency AB scores in the mixed avoidance task, alcohol versus nonalcohol: $r(58) = .087$, $p = .508$ and alcohol

Figure 2
Proportion of Trials With Break Frequency in the Avoidance Tasks



Note. The boxplots compare the proportion of trials showing attentional breaks (i.e., break frequency) in (a) the mixed and (b) predictable versions of the avoidance task for alcohol, nonalcohol, and nonappetitive distractors in patients with sAUD and CTL. Dots indicate individual data. sAUD = severe alcohol use disorder; CTL = control participants.

versus nonappetitive: $r(58) = .007$, $p = .958$, nor in the predictable avoidance task, alcohol versus nonalcohol: $r(54) = .193$, $p = .154$ and alcohol versus nonappetitive: $r(54) = .065$, $p = .633$.

AB–Craving

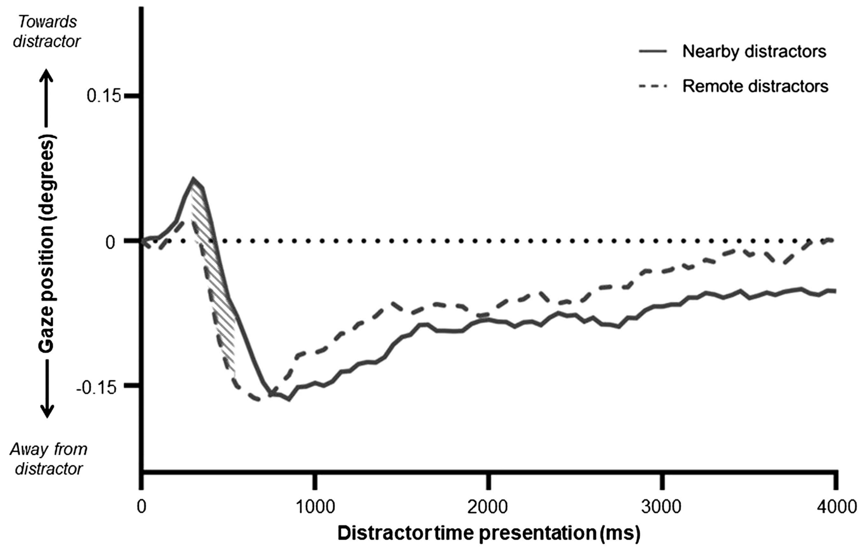
In the whole sample, craving correlated with dwell time AB score in the VPT, VAS: $r_s(58) = .301$, $p = .019$; ACQ: $r_s(58) = .285$, $p = .027$. It did not correlate with break frequency AB scores comparing alcohol with nonalcohol VAS: $r_s(58) = -.078$, $p = .554$; ACQ: $r_s(58) = -.141$, $p = .283$ or nonappetitive VAS: $r_s(58) = .011$, $p = .932$; ACQ: $r_s(58) = -.217$, $p = .095$ stimuli in the mixed avoidance task, nor with break frequency AB scores comparing alcohol with nonalcohol VAS: $r_s(54) = .147$, $p = .281$; ACQ: $r_s(54) = .073$, $p = .591$ or nonappetitive VAS: $r_s(54) = -.066$, $p = .630$; ACQ: $r_s(54) = -.171$, $p = .206$ stimuli in the predictable avoidance task.

Discussion

Dominant models consider alcohol-related AB as a core characteristic of sAUD (Wiers et al., 2007). However, little is known about the processes underlying AB. Capitalizing on an avoidance task using a gaze contingency procedure, we first investigated the compulsive versus reflective nature of AB by exploring how inhibitory control might influence its occurrence in patients with sAUD. We also manipulated the expectancy of stimuli type by presenting trials with alcohol, nonalcohol, and nonappetitive stimuli either in a randomized (mixed version) or sequential (predictable version) order. We then determined the convergent validity of this novel AB paradigm compared to the traditional VPT combined with reliable eye-tracking measures. Finally, we explored how craving could impact AB magnitude.

A first important result is that, in both versions of the avoidance task, we found no main effect of group and no interaction between group and stimulus type. This suggests that patients with sAUD do not present a global or alcohol-specific difficulty to inhibit saccades toward distractors when instructed to keep their attentional focus on a central target, and were hence able to correctly perform the task. Moreover, in the predictable version of the task, we showed that all participants were less likely to break their focus and attend to alcohol distractors compared to nonalcohol and nonappetitive ones. These findings are opposite to previous studies using this task in subclinical populations (Qureshi et al., 2019; Wilcockson & Pothos, 2015), as these earlier works found higher break frequencies toward alcohol and nonalcohol appetitive stimuli among heavy/problem drinkers. In contrast, we showed that both controls and patients with sAUD deliberately avoided alcohol-related stimuli once aware that they will be exclusively exposed to these stimuli. These findings are in line with our VPT results, indexing an avoidance AB for alcohol at later and more controlled attentional

Figure 3
Gaze Position of Fixational Eye Movements in the Mixed Avoidance Task



Note. The graph illustrates the gaze position (centered distance in degrees from target center) of fixational eye movements of the whole sample in the mixed version of the avoidance task, as a function of the time bins after distractor onset (one bin = 50 ms). The significant differences between nearby and remote distractors are indicated in gray stripes.

stages. Indeed, we observed shorter dwell times toward alcohol compared to nonalcohol stimuli in the VPT, especially among patients with sAUD. Patients with sAUD were also less likely to direct their gaze toward alcohol-related stimuli after a first fixation on nonalcohol stimuli. In contrast, this avoidance AB was not observed at the earlier and automatic processing stages (as indexed by first fixation location and duration measures). However, in line with previous results (Bollen et al., 2020, 2021), these measures presented a very low reliability, which was also observed for reaction time measures. Those behavioral measures are therefore unreliable and potentially misleading.

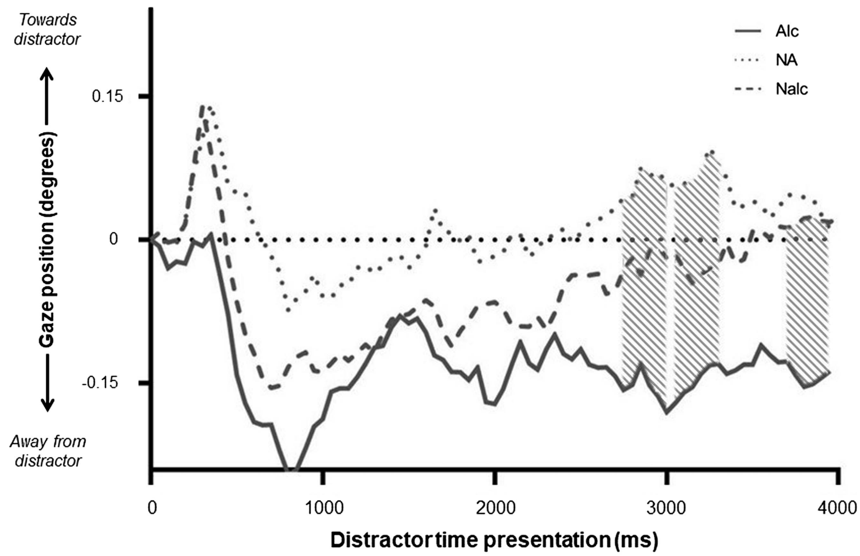
We also perfectly replicated previous findings regarding AB in sAUD (Bollen et al., 2021) by showing the presence of an avoidance AB for alcohol in sAUD through reliable eye-tracking indexes of later processes (i.e., dwell time, second fixation location). Although contradictory to dominant models (Franken, 2003), these results are not counterintuitive, since the presence of an avoidance AB pattern in detoxified patients under treatment could appear as a functional and adaptive process. Indeed, this population is usually characterized by negative (or ambivalent) evaluation of alcohol, low craving (only five patients reported craving higher than zero in the present study), and high motivation to stay abstinent (Field et al., 2016). Therefore, spontaneously avoiding processing alcohol-related stimuli might help them to regulate the negative affect associated with the exposure to those stimuli and would be even more easy to implement in clinical contexts in which the exposure to alcohol-related stimuli would not result in subsequent alcohol use.

Hence, we provide further experimental support for the recent proposal (Field et al., 2016) that AB is not a long-lasting and stable feature of sAUD but rather fluctuates alongside subjective evaluations of alcohol. While the avoidance AB was stronger in patients with sAUD than controls in the VPT, we did not find such significant group difference in the predictable version of the avoidance task. Altogether, we demonstrated that detoxified patients do not present a strong AB toward alcohol but rather exercise cognitive control over the visual exploration and processing of these stimuli, resulting in an avoidance AB. AB being a multidimensional construct, the VPT and avoidance task rely on distinct processes (as highlighted by the absence of correlations between the most reliable indexes of both tasks). On the one hand, the VPT measures the spontaneous trend to allocate attentional resources to specific stimuli. On the other hand, the avoidance task tests the ability to explicitly control attentional resources allocation.

In contrast with Qureshi et al. (2019), we observed higher break frequencies for nearby compared to remote distractors in both versions of the task. In their study, they argued that inhibitory processes of AB were affected by the ability to covertly process stimuli and that higher break frequencies for remote stimuli might be explained by the impossibility to recruit covert attention to process these stimuli without eye movements. Although central vision is indeed much more suited to fine discrimination than peripheral one, this latter argument is countered by many studies showing that peripheral vision still allows coarse discrimination such as object categorization, even at eccentricities up to 60°–70° (Boucart et al., 2013; D'Hondt et al., 2016; Thorpe et al., 2001). Moreover, our opposite findings rather suggest that stimuli which are easier to covertly process and categorize (i.e., nearly located stimuli) might be more distractive and thus more liable to attract gaze.

To clarify this point, we directly explored covert attention by measuring the gaze position of fixational eye movements when participants maintained their focus on the target object (Engbert, 2006; Laubrock et al., 2007; Lv et al., 2022). First, we showed that

Figure 4
Gaze Position of Fixational Eye Movements Among Patients in the Predictable Avoidance Task



Note. The graph illustrates the gaze position (centered distance in degrees from target center) of fixational eye movements of patients with sAUD in trials with nearby distractors of the predictable version of the avoidance task, as a function of the time bins after distractor onset (one bin = 50 ms). The significant differences between Alc Nalc and NA distractors are indicated in gray stripes. sAUD = severe alcohol use disorder; Alc = alcohol; Nalc = nonalcohol; NA = nonappetitive.

participants indeed recruit covert attention to process nearly located distractors in both versions of the task, as indexed by fixational eye movements getting closer to those stimuli compared to remote ones shortly after their onset. Second, we showed that fixational eye movements of patients with sAUD were positioned further away from alcohol distractors compared to nonappetitive distractors throughout the whole trial period in the predictable version of the task. In line with results on overt attention (i.e., break frequency in the predictable avoidance task, dwell time, and second fixation location in the VPT), these exploratory findings suggest that the avoidance AB of alcohol-related stimuli might also be indexed by covert attention. Future work should deepen these exploratory results by using paradigms directly dedicated to covert attention measurements, such as pupil light response, while manipulating the luminance of areas on the screen (e.g., Mathôt & Van der Stigchel, 2015; Salvaggio et al., 2022; Strauch et al., 2022). Indeed, the pupil size is not only modulated by the actual amount of light entering the eye but also reacts to the luminance of stimuli presented in peripheral vision, therefore revealing the locus of covert attention. Using pupillometry in addition to spatial eye movements research could thus allow to better appraise our understanding of covert attentional shifts in alcohol-related AB, especially in ambivalent drinkers that might attempt to override eye movements toward alcohol.

Importantly, the avoidance AB observed here appears specific to alcohol and does not generalize to other appetitive stimuli, since it was not observed for the nonalcohol appetitive beverages used in our tasks. This is in line with previous findings showing an avoidance AB for alcohol-related stimuli in patients with sAUD compared to nonalcohol appetitive stimuli (Bollen et al., 2021). However, it contrasts with Qureshi et al. (2019), who showed an AB generalized to all appetitive stimuli in subclinical problem drinkers. The specific alcohol-related AB, regardless of its direction, might thus only be found in clinical alcohol use disorder, while subclinical populations would still present a more diversified approach AB to various kinds of rewarding stimulations. Finally, subjective craving at testing time correlated with dwell time at the VPT but not with AB indexes in the avoidance task. While preliminary (as less than 20% of patients reported craving), this result suggests that craving might mostly influence AB in naturalistic settings when participants are allowed to watch alcohol stimuli during free exploration (as in the VPT). Conversely, the explicit instruction to inhibit eye movements toward alcohol (as in the avoidance task) would annihilate the influence of craving on AB measure. More globally, the low reported craving, together with other characteristics of our sAUD sample (i.e., negative evaluations of alcohol, interrupted alcohol consumption, high motivation to remain abstinent in the long term), leads to the proposal that the avoidance AB we report is an adaptive behavior, coherent with this population's current goals. Future work could use the avoidance task to compare patients with sAUD with or without subjective craving and ambivalent evaluation of alcohol, to better understand the role played by reflective abilities among individuals more prone to actually present approach AB toward alcohol. Indeed, as the instruction of the avoidance task was in line with our sample's spontaneous eye movements (i.e., tendency to avoid alcohol-related stimuli in the VPT without receiving explicit instructions to do so;

Bollen et al., 2021), our study did not explore the compulsive nature of AB in patients presenting difficulties to voluntary inhibit this AB.

Conclusion

We provide experimental support to the proposal that alcohol-related AB as evaluated by reliable eye-tracking measures is (a) not a long-lasting and stable characteristic of detoxified patients with sAUD and (b) underpinned by distinct attentional mechanisms that can be measured through different tasks and indexes. Indeed, we showed, through multiple experimental indexes, the robust presence of an avoidance AB specific to alcohol-related stimuli in patients with sAUD. This avoidance AB was underpinned by later and more controlled processes of AB in the VPT (as indexed by shorter dwell time and lower proportion of second fixation toward alcohol), but also by inhibitory processes (as indexed by lower break frequencies for alcohol distractors) and covert attention shifts (as indexed by fixational eye movements positioned further away from alcohol distractors) in the predictable avoidance task. Future studies should clarify the role played by inhibitory abilities in AB among patients with sAUD reporting positive or ambivalent motivational states for alcohol, and hence being more prone to present a compulsive AB toward alcohol.

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